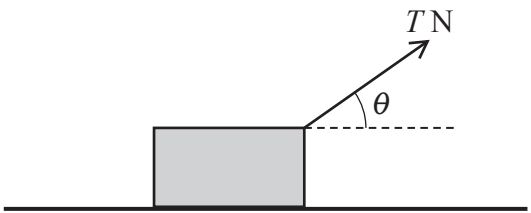
- The acceleration due to gravity is denoted by g m s^{-2}. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

16 A block of mass m kg rests on rough horizontal ground. The coefficient of friction between the block and the ground is μ. A force of magnitude T N is applied at an angle θ radians above the horizontal as shown in the diagram and the block slides without tilting or lifting.



(a) Show that the acceleration of the block is given by $\frac{T}{m}\cos\theta - \mu g + \frac{T}{m}\mu\sin\theta$. **[4]**

For a fixed value of T, the acceleration of the block depends on the value of θ. The acceleration has its greatest value when $\theta = \alpha$.

(b) Find an expression for α in terms of μ. **[3]**